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MTH 316 — Past Questions & Materials
by Renance

Course: MTH 316
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — MTH 316 Introduction To Mathematical Modeling 2020 2021 First Semester

Level 300 • First Semester • PastQuestion

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MARKING GUIDE

MTH 316

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
(FUPRE)

DEPARTMENT OF MATHEMATICS

2020/2021 FIRST SEMESTER EXAMINATION

MTH 316 - INTRODUCTION TO MATHEMATICAL MODELING

Instruction: Answer ANY THREE Questions

Time Allowed: 2Hours

Date: Friday 25th November, 2021.

- (1) (a) What is a Mathematical Modeling?
(b) Give any five area with problems that are tractable into mathematical formulation.
(c) Why Mathematical modeling?
(d) How do we classify any mathematical model?
[3, 10, 4, 8]Marks
- (2) (a) Describe the main activities in the process of mathematical modeling.
(b) Develop a model which will determine the rotational rate as a function of the station and the desired gravity. Use the model to determine what rotational rate will be required if the radius of the station is 250m and Earth surface gravity is desired? (take $g = 9.8m/s^2$)
[10, 15]Marks
- (3) (a) Consider the diagram below, state the basic assumptions required to

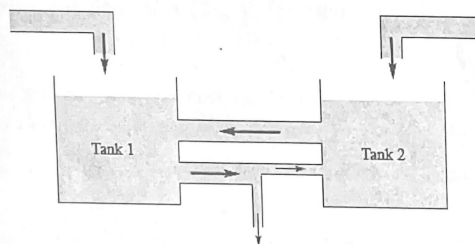


FIGURE 1. Mixing problem diagram

set up a system of equation to determine the quantity of contaminant/salt in each tank at any given time.

- (b) Two 1500litre tank are with salt water. Tank 1 contains 1000litres of water initially contains 100grams of salt dissolved in it and tank 2 contains 1200litres of water initially has 250grams of salt dissolved in it. Salt water with a concentration of $\frac{1}{2}$ grams/litre of salt enters tank 1 at the rate of 4litres/hour. Fresh water enters tank 2 at a rate of 7litres/hour. Through a connecting pipe water flows from tank 2 into tank 1 and 11litres/hour are drained out of the pipe (and hence leaving the system completely) and only 3litres/hour flows back into tank 2. Set up the system that will give the amount of salt in each tank at any given time.

[10, 15]Marks

Paper 2 of 2 — MTH 316, 2019

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